

Multiple Choice

1. **Answer:** C

Options: A) 1504 kPa; B) 1.5 MPa; C) 1.604 MPa; D) 1.4 MPa; E) 604 kPa

Note: Correct option: C - 1.604 MPa

2. **Answer:** B

Options: A) high and high; B) high and low; C) low and low; D) low and high

Note: Correct option: B - high and low

3. **Answer:** A

Options: A) $3/4$; B) $2/5$; C) $1/2$; D) $1/3$; E) $1/5$.

Note: Correct option: A - $3/4$.

4. **Answer:** D

Options: A) $E(X) = 3/2$, $\text{Var}(X) = 1/4$; B) $E(X) = 2$, $\text{Var}(X) = 1/3$; C) $E(X) = 5/3$, $\text{Var}(X) = 5/9$; D) $E(X) = 1$, $\text{Var}(X) = 2/3$; E) $E(X) = 3/4$, $\text{Var}(X) = 3/8$

Note: Correct option: D - $E(X) = 1$, $\text{Var}(X) = 2/3$

5. **Answer:** C

Options: A) star winding; B) delta winding; C) closed winding; D) parallel winding; E) zigzag winding

Note: Correct option: C - closed winding

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Hartley oscillator'.

Long Form Response

11. **Answer:** $n! / s^n + 1$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 1.90, 440×10^{-7} , 530×10^{-9} . *Explain why this is the correct answer and provide supporting evidence.*

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key